Research Project Checklist 1

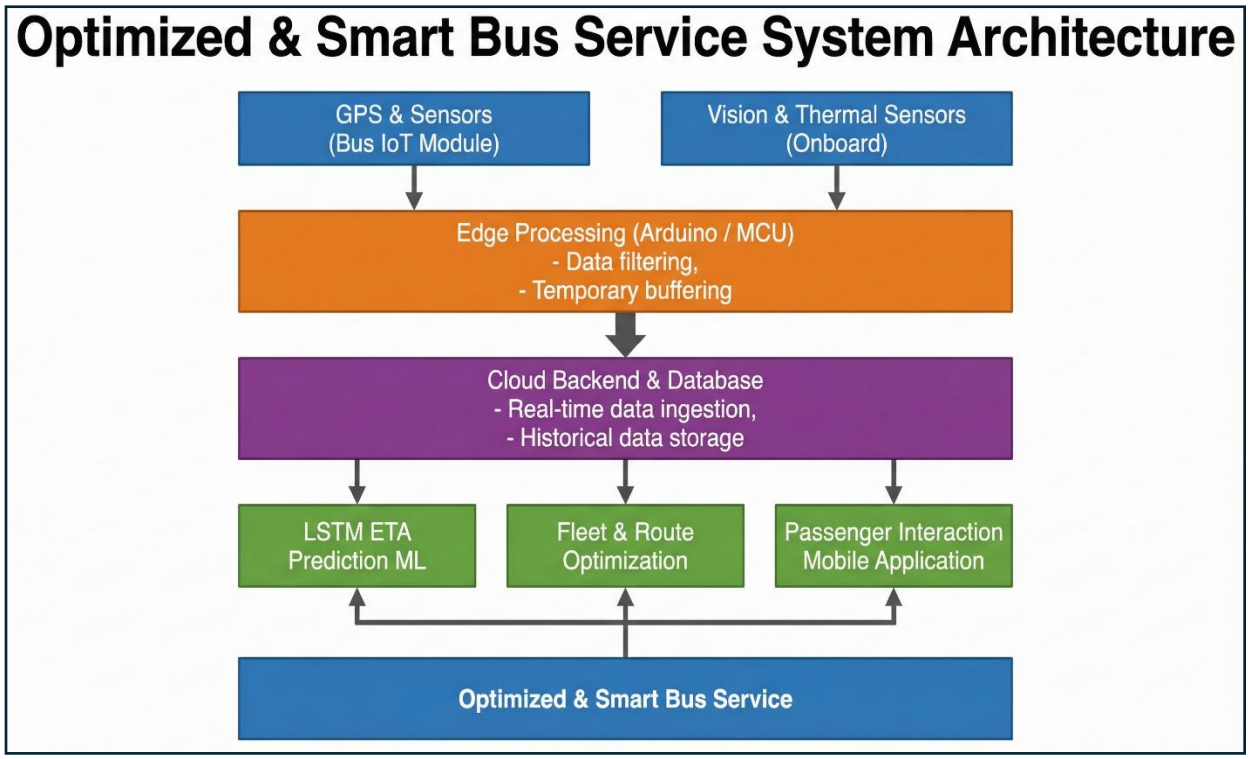
25-26J-267

GitHub Links

Frontend--- https://github.com/oshadha2k01/NextStop-Reaserach-frontend.git

Backend--- https://github.com/oshadha2k01/NextStop-Reaserach-backend.git

System Architecture



README.md

Smart Bus System for Real-Time Passenger Management

Project Overview

The *Smart Bus System for Real-Time Passenger Management* is an IoT-driven, intelligent public transport solution designed to improve efficiency, accessibility, and passenger experience in Sri Lanka's bus network. The system integrates *real-time tracking, passenger occupancy monitoring, mobile interaction, and AI-driven fleet optimization* into a single smart ecosystem.

This project is developed as part of the *IT4010 Research Project – 2025 (July Intake)* under the *Information Technology (IT)* specialization.

Main Objectives

* Provide *real-time bus location and accurate ETA predictions*

* Reduce *overcrowding* by monitoring passenger occupancy

* Enable *inclusive and accessible passenger interaction*

* Optimize *routes and fleet usage* using AI-driven analytics

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System Architecture

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| GPS & Sensors | | Vision & Thermal |

| (Bus IoT Module) | | Sensors (Onboard) |

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| Edge Processing (Arduino / MCU) |
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```
| - Data filtering |
| - Temporary buffering |
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|
v
+----------------------------------+
| Cloud Backend & Database |
| - Real-time data ingestion |
| - Historical data storage |
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| LSTM ETA | | Fleet & Route | | Passenger Interaction|
| Prediction ML | | Optimization | | Mobile Application |
+---------------+ +----------------+ +----------------------+
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v
Optimized & Smart Bus Service
```

Core Functional Modules

IoT-Based Real-Time Bus Tracking & ETA Prediction

Owner: Weerakoon W.M.D.P (IT22131560)

Functionality:

* GPS-based real-time bus tracking using Arduino

* Hybrid *GSM + Wi-Fi* communication

* Cloud-based data pipeline

* *LSTM machine learning model* for accurate ETA prediction

Novelty:

* Context-aware ETA prediction using traffic, weather, and event data

* Low-cost and scalable IoT deployment for developing regions

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Vision-Based Passenger Counting & Occupancy Monitoring

Owner: Abeysekara W.R.G.M (IT22271150)

Functionality:

* Real-time passenger counting using cctv cameras for both exit and entrance doors.

* Embedded deep learning models (Yolo12) for human detecting.

* Privacy-preserving anonymous detection

Novelty:

* Realtime calculation share with mobile and desktop users.

* No GPU dependency – optimized for embedded systems

Mobile-Based Passenger Interaction & Accessibility System

Owner: Chandrasekara C.M.P.V (IT22286246)

Functionality:

* Mobile app for real-time bus updates

* Stop request and seat availability alerts

* Multilingual, audio, and vibration support

Novelty:

* Inclusive design for differently-abled users

* Smart travel suggestions based on live occupancy

Central Fleet Management & Route Optimization Engine

Owner: Pathiraja H.P.M.O.N (IT22230492)

Functionality

- AI/ML-based prediction model using real-time IoT data to estimate bus arrival time at the passenger's selected destination before onboarding.

- Dynamic ticket price calculation based on the start and destination locations, using travel distance calculated before the journey begins.

- Live bus tracking with real-time alerts and warnings generated from passenger complaints related to driver behavior.

Novelty

- Pre-onboarding AI/ML arrival-time prediction that allows passengers to know the expected travel duration in advance.

- Transparent ticket pricing to reduce corruption and improve driver behavior through real-time passenger feedback and notifications.

Project Team

| Name | Registration Number | Responsibility |

| --- | --- | --- |

| Weerakoon W.M.D.P | IT22131560 | IoT Tracking & ETA Prediction |

| Pathiraja H.P.M.O.N | IT22230492 | Fleet Management & Optimization |

| Chandrasekara C.M.P.V | IT22286246 | Mobile App & Accessibility |

| Abeysekara W.R.G.M | IT22271150 | Passenger Counting & Vision |

Technologies Used

* *Hardware:* Arduino, GPS, Thermal Sensors, Cameras

* *Networking:* GSM, Wi-Fi

* *Backend:* Cloud Database, REST APIs

* *AI/ML:* LSTM, Tiny-YOLOv3, MobileNet

* *Mobile:* Flutter / Android

* *Analytics:* Python, Data Analytics

Conclusion

This Smart Bus System introduces a *scalable, intelligent, and inclusive public transport solution* tailored for Sri Lanka. By integrating IoT, AI, computer vision, and mobile technologies, the system enhances reliability, reduces congestion, and improves daily commuting experiences for millions of passengers.

This README.md is prepared for academic and project documentation purposes (IT4010 Research Project – 2025)

Research Project Checklist 2

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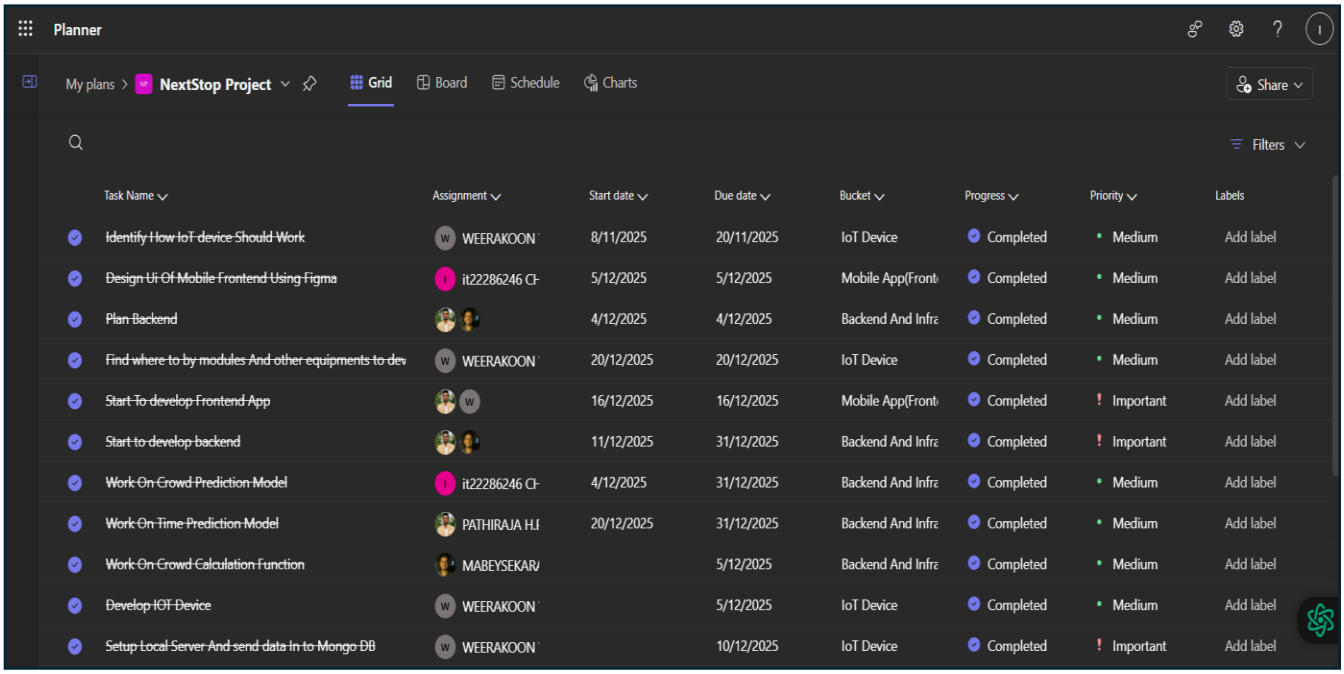
Project Management Tool :- Microsoft Planner

Grid View

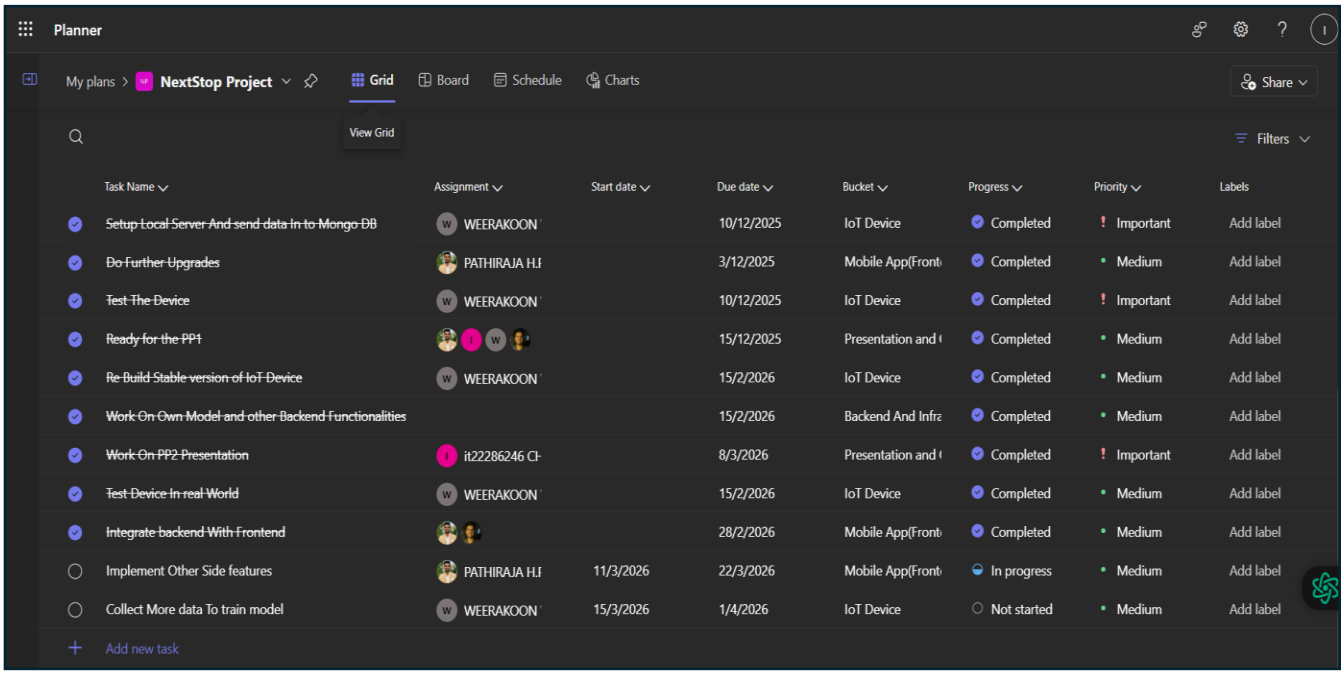


Planner

My plans > NextStop Project | Grid | Board | Schedule | Charts | Share

Filters

Task Name	Assignment	Start date	Due date	Bucket	Progress	Priority	Labels
~~Identify How IoT device Should Work~~	WEERAKOON	8/11/2025	20/11/2025	IoT Device	Completed	Medium	Add label
~~Design Ui Of Mobile Frontend Using Figma~~	it22286246 CH	5/12/2025	5/12/2025	Mobile App(Front	Completed	Medium	Add label
~~Plan Backend~~		4/12/2025	4/12/2025	Backend And Infra	Completed	Medium	Add label
~~Find where to by modules And other equipments to dev~~	WEERAKOON	20/12/2025	20/12/2025	IoT Device	Completed	Medium	Add label
~~Start To develop Frontend App~~		16/12/2025	16/12/2025	Mobile App(Front	Completed	Important	Add label
~~Start to develop backend~~		11/12/2025	31/12/2025	Backend And Infra	Completed	Important	Add label
~~Work On Crowd Prediction Model~~	it22286246 CH	4/12/2025	31/12/2025	Backend And Infra	Completed	Medium	Add label
~~Work On Time Prediction Model~~	PATHIRAJA H.I	20/12/2025	31/12/2025	Backend And Infra	Completed	Medium	Add label
~~Work On Crowd Calculation Function~~	MABEYSEKARA		5/12/2025	Backend And Infra	Completed	Medium	Add label
~~Develop IOT Device~~	WEERAKOON		5/12/2025	IoT Device	Completed	Medium	Add label
~~Setup Local Server And send data In to Mongo DB~~	WEERAKOON		10/12/2025	IoT Device	Completed	Important	Add label



Planner

My plans > NextStop Project | Grid | Board | Schedule | Charts | Share

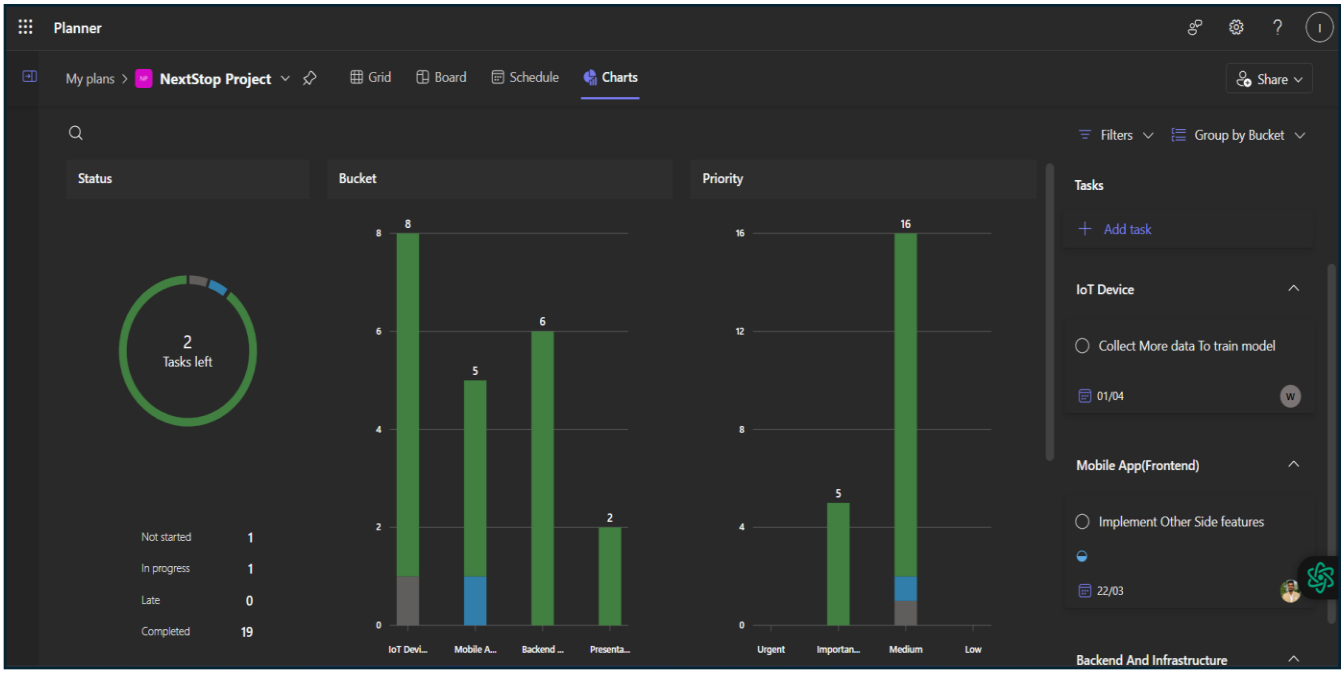
View Grid

Filters

Task Name	Assignment	Start date	Due date	Bucket	Progress	Priority	Labels
~~Setup Local Server And send data In to Mongo DB~~	WEERAKOON		10/12/2025	IoT Device	Completed	Important	Add label
~~Do Further Upgrades~~	PATHIRAJA H.I		3/12/2025	Mobile App(Front	Completed	Medium	Add label
~~Test The Device~~	WEERAKOON		10/12/2025	IoT Device	Completed	Important	Add label
~~Ready for the PP1~~			15/12/2025	Presentation and	Completed	Medium	Add label
~~Re Build Stable version of IoT Device~~	WEERAKOON		15/2/2026	IoT Device	Completed	Medium	Add label
~~Work On Own Model and other Backend Functionalities~~			15/2/2026	Backend And Infra	Completed	Medium	Add label
~~Work On PP2 Presentation~~	it22286246 CH		8/3/2026	Presentation and	Completed	Important	Add label
~~Test Device In real World~~	WEERAKOON		15/2/2026	IoT Device	Completed	Medium	Add label
~~Integrate backend With Frontend~~			28/2/2026	Mobile App(Front	Completed	Medium	Add label
Implement Other Side features	PATHIRAJA H.I	11/3/2026	22/3/2026	Mobile App(Front	In progress	Medium	Add label
Collect More data To train model	WEERAKOON	15/3/2026	1/4/2026	IoT Device	Not started	Medium	Add label

+ Add new task

Charts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My plans
NextStop Project
Grid
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Schedule
Charts
Share
Filters
Group by Bucket
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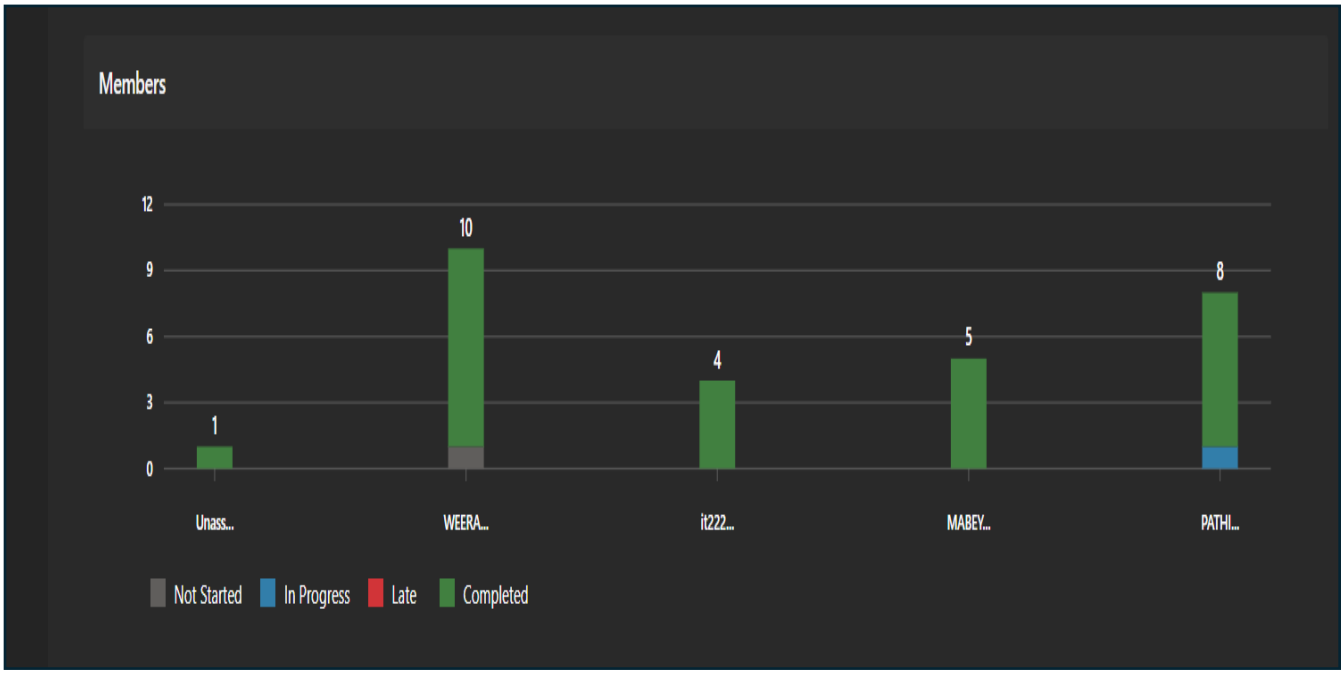
Mobile App(Frontend)
Implement Other Side features
22/03
Backend And Infrastructure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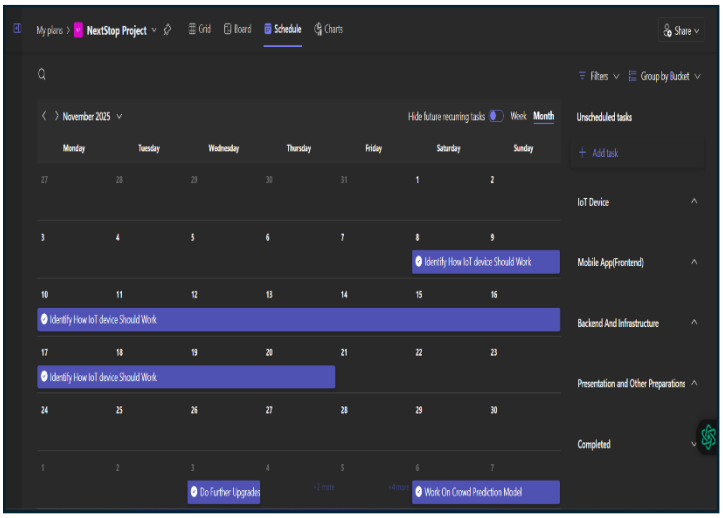
Schedule


My plans
NextStop Project
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Schedule
Charts
Share
Filters
Group by Bucket
November 2025
Hide future recurring tasks
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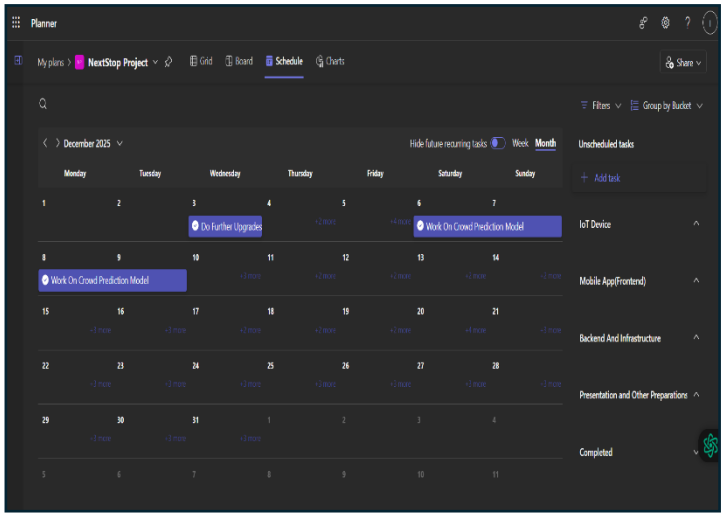
Unscheduled tasks
Add task
IoT Device
Mobile App(Frontend)
Backend And Infrastructure
Presentation and Other Preparations
Completed
Identify How IoT device Should Work
Identify How IoT device Should Work
Identify How IoT device Should Work
Do Further Upgrades
Work On Crowd Prediction Model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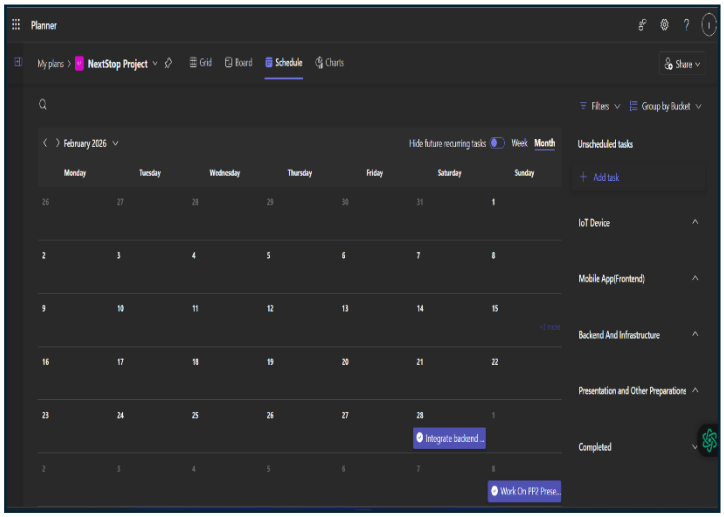
IoT Device
Mobile App(Frontend)
Backend And Infrastructure
Presentation and Other Preparations
Completed
Do Further Upgrades
Work On Crowd Prediction Model
Work On Crowd Prediction Model


Planner
My plans
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Filters
Group by Bucket
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Hide future recurring tasks
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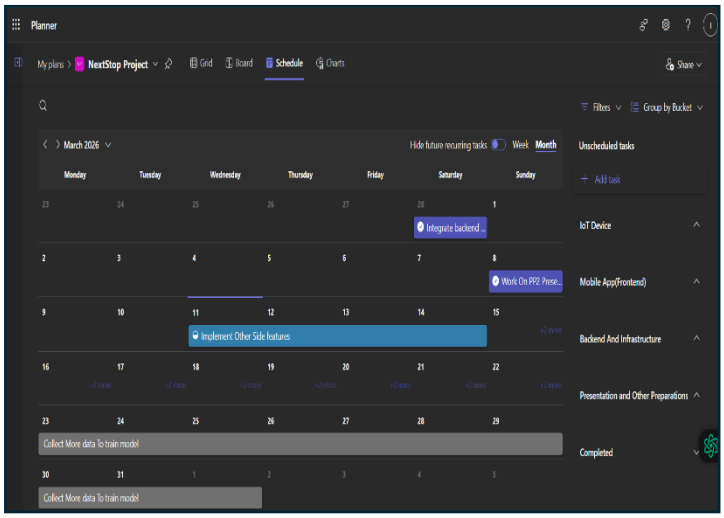
Unscheduled tasks
Add task
IoT Device
Mobile App(Frontend)
Backend And Infrastructure
Presentation and Other Preparations
Completed
Integrate backend ...
Work On PP2 Prese...


Planner
My plans
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Group by Bucket
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Unscheduled tasks
Add task
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